

# Monica Munnangi

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## RESEARCH INTERESTS

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LLMs, Foundation Model Reliability, Robustness & Evaluation, Multi-modality, Retrieval-Augmented Generation (RAG), Clinical NLP, AI Safety & Alignment, Large-scale Pretraining, Human-AI Interaction

## EDUCATION

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- **Northeastern University** Boston, MA  
*PhD in Computer Science* *Sep 2021 - Present*
  - **Advisor:** Prof. Saiph Savage
- **University of Massachusetts Amherst** Amherst, MA  
*Master of Science in Computer Science* *Sep 2018 - May 2020*
  - **Research:** Information Fusion Lab, advised by Prof. Madalina Fiterau
- **Vellore Institute of Technology** Chennai, India  
*Bachelor of Technology in Computer Science and Engineering* *Jul 2014 - May 2018*

## EXPERIENCE

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- **Northeastern University** Boston, MA  
*Graduate Research Assistant* *Sep 2021 - Present*
  - **Designing novel evaluation frameworks and multi-turn datasets** to rigorously benchmark LLM factuality, domain adaptation, and real-world reliability.
  - **Engineered scalable training and evaluation pipelines** in PyTorch, successfully sharding large foundation models (e.g., 70B parameters) across GPU clusters using DeepSpeed and PyTorch Lightning.
  - **Evaluated instruction sensitivity** of clinical-tuned models, published at **BioNLP at ACL, 2024**.
- **Stanford University** Palo Alto, CA  
*Research Intern - Shah Lab* *May 2024 - Aug 2024*
  - **Developed and open-sourced** a comprehensive dataset and entailment-based evaluation framework to detect critical omissions and hallucinations in clinical LLMs.
  - **Built automated pipelines** for large-scale data curation and factuality evaluation, shifting away from rigid metrics (like ROUGE) to atomic-fact decomposition for safe AI deployment.
  - **Published FactEHR:** dataset for evaluating factual accuracy in clinical LLMs, addressing critical reliability challenges for AI safety applications. Accepted to **Machine Learning for Healthcare, 2025**.
- **Allen Institute for AI** Seattle, WA  
*Research Intern - Semantic Scholar* *May 2023 - Aug 2023*
  - **Built a framework** to improve the performance of LLMs with knowledge augmentation in biomedical and scientific domain.
  - **Improved performance** over state-of-the-art models on biomedical IE with LLMs, we present ablations to test the understanding of the models. Work published at **NAACL, 2024**.
- **GE Healthcare** Waukesha, WI  
*Data Scientist Intern* *May 2019 - Aug 2019*
  - **Developed an NN** to identify the presence of a chest tube in an Pneumothorax patient's X-Ray, trained the model on 8000 images and fine tuned on the pre-trained VGG architecture.
  - **Architected and integrated** a SOTA-surpassing model (95% accuracy) into production environments, utilizing PyTorch to automate high-risk patient stratification and optimize radiologist diagnostic pipelines. Work accepted as a poster at **European Congress of Radiology, 2020**.
- **DoctorC (Simplify Wellness Pvt. Ltd)** Hyderabad, IN  
*Software Developer Intern* *Jan 2018 - Apr 2018*
  - **Engineered high-performance** mobile architectures using React Native and Xcode, optimizing frontend-to-backend data synchronization via RESTful API services to achieve a 25% increase in system usability.

## PUBLICATIONS

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- **ThReadMed-QA: A Multi-Turn Medical Dialogue Benchmark from Real Patient Questions**  
M. Munnangi, Saiph Savage  
*Under Review*
- **FactEHR: A Dataset for Evaluating Factuality in Clinical Notes Using LLMs**  
Monica Munnangi\*, Akshay Swaminathan\*, Jason Alan Fries\*, Jenelle Jindal, Sanjana Narayanan, Ivan Lopez, Lucia Tu, Philip Chung, Jesutofunmi A. Omiye, Mehr Kashyap, Nigam Shah  
*Accepted to Machine Learning for Healthcare 2025*
- **Open (Clinical) LLMs are Sensitive to Instruction Phrasings**  
Alberto Mario Ceballos-Arroyo\*, Monica Munnangi\*, Jiuding Sun, Karen Zhang, Jered McInerney, Byron C. Wallace, Silvio Amir  
*Accepted to BioNLP at ACL 2024*
- **On-the-fly Definition Augmentation of LLMs for Biomedical NER**  
Monica Munnangi, Sergey Feldman, Byron Wallace, Silvio Amir, Tom Hope, Aakanksha Naik  
*Accepted to NAACL 2024*
- **Chest Tube Detection on Chest X-Ray Images Using Convolutional Deep Neural Network**  
Khaled Younis\*, Yash Shah\*, Monica Munnangi\*, Katelyn Nye, Gireesha Rao, John M. Sabol  
*Accepted as a poster at European Congress of Radiology, Vienna 2020*

## TEACHING/ ADVISING EXPERIENCE

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- Teaching Assistant for **HCI seminar** and assisted Prof. Saiph Savage at Northeastern University in Spring 2026 semester.
- Teaching Assistant for **Unsupervised Machine Learning and Data Mining** and assisted Prof. Pavlu Virgil at Northeastern University in Spring 2023 semester.
- Teaching Assistant for **Unsupervised Data Mining** and assisted Prof. Pavlu Virgil at Northeastern University in Fall 2022 semester.
- Co-advised a cohort of graduate students for a project titled Naik, A. et al. **Leveraging knowledge distillation for efficient on-device deployment of deep learning models in medical imaging** published in Society for Imaging Informatics in MCMI in Medical Imaging, Nov 2020.
- Teaching Assistant for the course **Database Management Systems** and assisted Prof. Muralidhar A. at Vellore Institute of Technology in the Fall 2017 semester.

## ACADEMIC SERVICE AND ACHIEVEMENTS

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- Co-chair for Conference on Health, Inference, and Learning 2023 and 2024
- Program Committee member for Human-centered LLMs workshop, ACL 2024
- Reviewer: NeurIPS 2022, AAAI 2022, ML4H 2020, 2021, 2022, 2023, 2024, 2025, MLHC 2025, 2026
- Program Committee member for GenAI4Health workshop, NeurIPS 2025

## SKILLS

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- Python, PyTorch, C++, Sklearn, Numpy, LangChain, Git, R, distributed training / HPC (e.g., Slurm), Docker, DeepSpeed, vLLM, HTML, CSS